microeconomics_2_L3

Subject: #microeconomics

Topic: Monopoly

Definition. A **monopoly** is a market structure where a single firm is the only seller of a product.

Monopoly Pricing Instruments:

- **Linear pricing:** the monopolist can charge a single per-unit price p.
  Think of this as the "normal" way of selling — one price tag per unit.
- **Two-part tariff:** the monopolist can charge a fixed fee plus a per-unit price.
  Like a membership fee at Costco plus the price of groceries inside.
- **Price discrimination:** charge different prices to different consumers (3rd degree) or for different quantities (2nd degree).
  Examples: student discounts (different buyers) or bulk discounts (different quantities).
- **Bundling:** sell multiple goods together as a package.
  E.g., Microsoft Office sells Word, Excel, and PowerPoint as one bundle instead of separately.

Causes of Monopoly

Two key drivers:

1. **Technology:** high fixed costs and large minimum efficient scale (AC^*).
   If it's very expensive to start producing, and only when producing at a large scale does the cost per unit fall low enough, only one big producer will survive profitably.
2. **Market size:** small demand relative to AC^*.
   If there just isn't enough demand to support multiple producers, one firm ends up serving the whole market.

> **Note**
>
> If only one or two firms can profitably operate in the long run due to high AC^* and small market size, a monopoly is likely to arise.

Linear Pricing Monopoly

Suppose that:

- The monopolist charges a uniform per-unit price p (linear pricing).
- Quantity sold is given by market demand: $q = D(p)$.
- Profit is: $\pi(p) = p \cdot D(p) - C(D(p))$.

Or, equivalently:

$$\max_{q \geq 0} \pi(q) = R(q) - C(q)$$

where $R(q) = p(q) \cdot q$ and $p(q) = D^{-1}(q)$ is the inverse demand.

Tip

The monopolist chooses **how much to sell**, knowing that this will affect the price they can charge. Unlike in perfect competition (where firms take the price as given), the monopolist controls price indirectly through its output choice.

Profit Maximization (FOC and SOC)

- **Marginal revenue:**

$$MR(q) = R'(q) = p(q) + p'(q) \cdot q$$

  *When a monopolist sells one more unit, it earns extra revenue at the current price ($p(q)$), **but** it also has to lower the price slightly for all units ($p'(q) \cdot q$). That's why MR is always below price.*

- **First-order condition (interior optimum):**

$$MR(q^m) = MC(q^m)$$

  The "sweet spot": the monopolist produces where the gain from selling one more unit (MR) equals the extra cost of making it (MC).

- **Second-order condition:**

$$MR'(q^m) - MC'(q^m) \leq 0$$

  This just checks that we really found a maximum (not a minimum). It says revenue should be flattening or falling faster than costs are rising.

- **Shutdown condition:**

$$\pi(q^m) \geq \pi(0) \quad \text{or} \quad p^m \geq AVC(q^m) + \frac{F - S}{q^m}$$

  The monopolist stays in business as long as it makes more money by producing than by shutting down. Importantly, it can survive at lower prices than a competitive firm, because it has pricing power.

Class Example 4:

Inverse demand:

$$p(q) = K - q, \quad K > 0$$

Firm total cost:

$$C(q) = \begin{cases} 12 & q = 0, \\ 16 + q^2 & q > 0. \end{cases}$$

Find the optimal monopoly quantity–price pair for $K = 4$ and $K = 8$.

Solution:

For $q > 0$ revenue and profit:

$$R(q) = (K - q)q = Kq - q^2,$$

$$\pi(q) = R(q) - C(q) = Kq - q^2 - (16 + q^2) = Kq - 2q^2 - 16.$$

First-order condition (FOC):

$$\pi'(q) = K - 4q = 0 \quad \Rightarrow \quad q^* = \tfrac{K}{4}.$$

Second-order condition (SOC):

$$\pi''(q) = -4 < 0,$$

so the critical point is a maximum for $q > 0$.

Price at the interior solution:

$$p^* = K - q^* = K - \tfrac{K}{4} = \tfrac{3K}{4}.$$

Profit at the interior solution:

$$\pi(q^*) = \tfrac{K^2}{8} - 16.$$

Shutdown check.
If the firm produces $q > 0$ it gets $\pi(q^*)$. If it shuts down ($q = 0$) profit is $\pi(0) = -12$. The firm produces only if

$$\pi(q^*) \geq \pi(0) \quad \Longleftrightarrow \quad \tfrac{K^2}{8} - 16 \geq -12,$$

$$\Longleftrightarrow \quad \tfrac{K^2}{8} \geq 4 \quad \Longleftrightarrow \quad K^2 \geq 32 \quad \Longleftrightarrow \quad K \geq \sqrt{32} \approx 5.657.$$

So:

- If $K < \sqrt{32}$ the best choice is to shut down: $q^m = 0$.
- If $K \geq \sqrt{32}$ the interior solution $q^* = K/4$, $p^* = 3K/4$ is chosen.

Case $K = 4$

Interior candidate:

$$q^* = \tfrac{4}{4} = 1, \qquad p^* = \tfrac{3 \cdot 4}{4} = 3.$$

Profit at that point:

$$\pi(1) = \tfrac{4^2}{8} - 16 = 2 - 16 = -14.$$

Compare to shutdown profit $\pi(0) = -12$. Since $-14 < -12$, producing is worse.

Thus ($K = 4$):

- Optimal: shut down, $q^m = 0$.
- Profit: $\pi(0) = -12$.

Case $K = 8$

Interior candidate:

$$q^* = \tfrac{8}{4} = 2, \qquad p^* = \tfrac{3 \cdot 8}{4} = 6.$$

Profit at that point:

$$\pi(2) = \frac{8^2}{8} - 16 = 8 - 16 = -8.$$

Compare to shutdown profit $\pi(0) = -12$. Since $-8 > -12$, producing is better.

Thus $(K = 8)$:

- Optimal: $q^m = 2$, $p^m = 6$.
- Profit: $\pi(2) = -8$.

Price Elasticity and Lerner Index

- **Price elasticity of demand**:

$$\epsilon(p) = \frac{p \cdot D'(p)}{D(p)} \quad \text{or} \quad \epsilon(q) = \frac{p(q)}{p'(q) \cdot q}$$

- **Lerner Index of Market Power (LIMP)**:

$$\frac{p^m - MC(q^m)}{p^m} = -\frac{1}{\epsilon(q^m)}$$

  - LIMP ∈ [0,1]; higher values indicate more market power
  - A competitive firm has LIMP = 0
  - A monopolist never operates where demand is inelastic ($|\epsilon| < 1$)

Sources:

Additional materials:

Review questions:

1. What is the defining feature of a monopoly?

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2. What is linear pricing under monopoly?

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3. What is a two-part tariff?

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4. How does second-degree price discrimination differ from third-degree price discrimination?

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5. What are the two main causes of monopoly?

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6. In the monopoly profit maximization problem, how is revenue $R(q)$ defined in terms of the inverse demand $p(q)$?

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7. Why is marginal revenue (MR) always below the demand curve (price) for a monopolist?

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8. What condition characterizes the monopolist's optimal output q^m?

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9. What does the shutdown condition for a monopolist state?

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10. What is the Lerner Index of Market Power and how is it related to elasticity?
